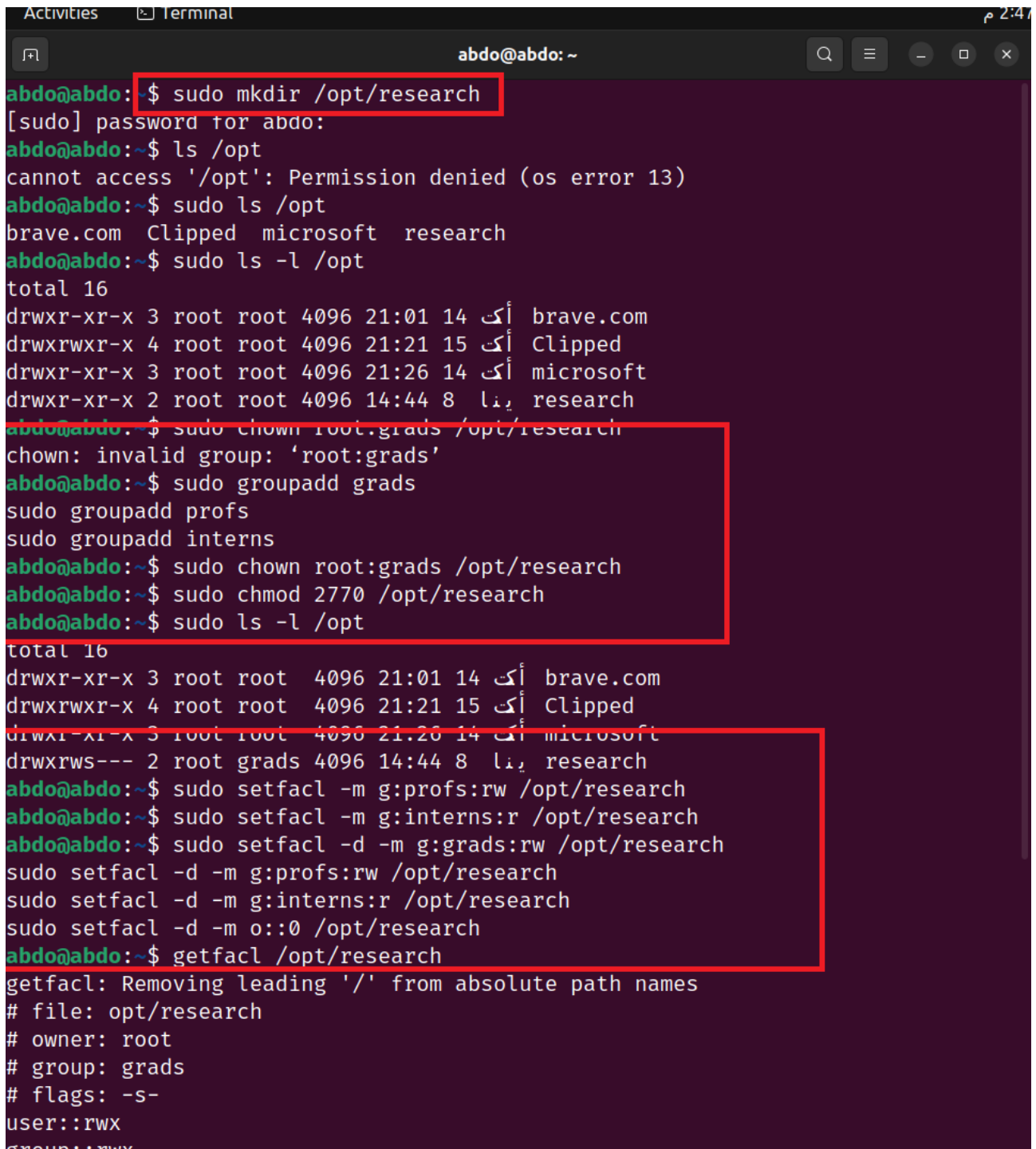


Lab_1_part_II

16. Grant your colleagues a collective directory called /opt/research, where they can store generated research results. Only members of group profs and grads should be able to create new files in the directory, and new file should have the following properties: the directory should be owned by root new files should be group owned by group grads group profs should automatically have read/write access to new files group interns should automatically have read only access to new files other users should not be able to access the directory and its contents at all.



```
abdo@abdo: ~  
abdo@abdo:~$ sudo mkdir /opt/research  
[sudo] password for abdo:  
abdo@abdo:~$ ls /opt  
cannot access '/opt': Permission denied (os error 13)  
abdo@abdo:~$ sudo ls /opt  
brave.com Clipped microsoft research  
abdo@abdo:~$ sudo ls -l /opt  
total 16  
drwxr-xr-x 3 root root 4096 21:01 14 أكت brave.com  
drwxrwxr-x 4 root root 4096 21:21 15 أكت Clipped  
drwxr-xr-x 3 root root 4096 21:26 14 أكت microsoft  
drwxr-xr-x 2 root root 4096 14:44 8 رينا research  
abdo@abdo:~$ sudo chown root:grads /opt/research  
chown: invalid group: 'root:grads'  
abdo@abdo:~$ sudo groupadd grads  
sudo groupadd profs  
sudo groupadd interns  
abdo@abdo:~$ sudo chown root:grads /opt/research  
abdo@abdo:~$ sudo chmod 2770 /opt/research  
abdo@abdo:~$ sudo ls -l /opt  
total 16  
drwxr-xr-x 3 root root 4096 21:01 14 أكت brave.com  
drwxrwxr-x 4 root root 4096 21:21 15 أكت Clipped  
drwxr-xr-x 3 root root 4096 21:26 14 أكت microsoft  
drwxrws--- 2 root grads 4096 14:44 8 رينا research  
abdo@abdo:~$ sudo setfacl -m g:profs:rw /opt/research  
abdo@abdo:~$ sudo setfacl -m g:interns:r /opt/research  
abdo@abdo:~$ sudo setfacl -d -m g:grads:rw /opt/research  
sudo setfacl -d -m g:profs:rw /opt/research  
sudo setfacl -d -m g:interns:r /opt/research  
sudo setfacl -d -m o::0 /opt/research  
abdo@abdo:~$ getfacl /opt/research  
getfacl: Removing leading '/' from absolute path names  
# file: opt/research  
# owner: root  
# group: grads  
# flags: -s-  
user::rwx  
group::rwx
```

17. Display your MAC address by 2 different ways.

```
abdo@abdo: ~  
abdo@abdo:~$ ip link show enp2s0  
2: enp2s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group default qlen 1000  
    link/ether 3c:2c:30:f4:ef:e3 brd ff:ff:ff:ff:ff:ff  
abdo@abdo:~$ ifconfig enp2s0  
enp2s0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether 3c:2c:30:f4:ef:e3 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
abdo@abdo:~$
```

18. Display the network settings of all active interfaces.

```
TX packets 0 bytes 0 (0.0 B)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
abdo@abdo:~$ nmcli device show  
GENERAL.DEVICE: wlo1  
GENERAL.TYPE: wifi  
GENERAL.HWADDR: D8:F2:CA:72:17:35  
GENERAL.MTU: 1500  
GENERAL.STATE: 100 (connected)  
GENERAL.CONNECTION: abdo 1  
GENERAL.CON-PATH: /org/freedesktop/NetworkManager/Activ>  
IP4.ADDRESS[1]: 192.168.1.41/24  
IP4.GATEWAY: 192.168.1.1  
IP4.ROUTE[1]: dst = 192.168.1.0/24, nh = 0.0.0.0, m>  
IP4.ROUTE[2]: dst = 169.254.0.0/16, nh = 0.0.0.0, m>  
IP4.ROUTE[3]: dst = 0.0.0.0/0, nh = 192.168.1.1, mt>  
IP4.DNS[1]: 192.168.1.1  
IP6.ADDRESS[1]: fe80::1949:c77c:1ce6:72ad/64  
IP6.GATEWAY: --  
IP6.ROUTE[1]: dst = fe80::/64, nh = ::, mt = 1024
```

```
abdo@abdo: ~  
abdo@abdo:~$ ip addr show up  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: enp2s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000  
    link/ether 3c:2c:30:f4:ef:e3 brd ff:ff:ff:ff:ff:ff  
3: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000  
    link/ether d8:f2:ca:72:17:35 brd ff:ff:ff:ff:ff:ff  
    altname wlp0s20f3  
    inet 192.168.1.41/24 brd 192.168.1.255 scope global dynamic noprefixroute wlo1  
        valid_lft 84133sec preferred_lft 84133sec  
    inet6 fe80::1949:c77c:1ce6:72ad/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
abdo@abdo:~$
```

19. Display the network setting of all interfaces both active inactive.

```
abdo@abdo: ~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp2s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 3c:2c:30:f4:ef:e3 brd ff:ff:ff:ff:ff:ff
3: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether d8:f2:ca:72:17:35 brd ff:ff:ff:ff:ff:ff
    altname wlp0s20f3
    inet 192.168.1.41/24 brd 192.168.1.255 scope global dynamic noprefixroute wlo1
        valid_lft 84088sec preferred_lft 84088sec
    inet6 fe80::1949:c77c:1ce6:72ad/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
abdo@abdo: ~$ ifconfig -a
enp2s0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 3c:2c:30:f4:ef:e3 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1520 bytes 212782 (212.7 KB)
```

20. Bring your interface down.
21. Configure your network card to have static IP.
22. Bring your interface up.

```
abdo@abdo: ~  
$ nmcli device status  
DEVICE      TYPE      STATE      CONNECTION  
wlo1         wifi      connected  abdo 1  
p2p-dev-wlo1 wifi-p2p   disconnected --  
enp2s0       ethernet  unavailable --  
lo           loopback  unmanaged  --  
$ nmcli con down "abdo 1"  
Connection 'abdo 1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)  
$ nmcli con mod "abdo 1" ipv4.addresses 192.168.1.100/24  
nmcli con mod "abdo 1" ipv4.gateway 192.168.1.1  
nmcli con mod "abdo 1" ipv4.dns "8.8.8.8 8.8.4.4"  
nmcli con mod "abdo 1" ipv4.method manual  
$ nmcli con up "abdo 1"  
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)  
$ ~
```

23. Verify your network setting using ifconfig command

```
abdo@abdo: ~  
$ nmcli device status  
DEVICE      TYPE      STATE      CONNECTION  
wlo1         wifi      connected  abdo 1  
dummy0       dummy     connected  lab-net  
p2p-dev-wlo1 wifi-p2p   disconnected --  
enp2s0       ethernet  unavailable --  
lo           loopback  unmanaged  --  
$ ifconfig dummy0  
dummy0: flags=195<UP,BROADCAST,RUNNING,NOARP> mtu 1500  
ether 3a:4f:eb:62:41:5b txqueuelen 1000 (Ethernet)  
RX packets 0 bytes 0 (0.0 B)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 0 bytes 0 (0.0 B)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
$
```

24. Configure your network card to have dynamic IP using network manager command.

```
abdo@abdo: ~  
abdo@abdo:~$ nmcli con mod lab-net ipv4.method auto  
abdo@abdo:~$ nmcli con up lab-net  
Error: Connection activation failed: IP configuration could not be reserved (no available address, timeout, etc.)  
Hint: use 'journalctl -xe NM_CONNECTION=3cf1b663-05f8-4741-a297-af18ceebf9e4 + NM_DEVICE=dummy0' to get more details.  
abdo@abdo:~$
```

25. Check using ifconfig then check its configuration file.

```
abdo@abdo: ~  
abdo@abdo:~$ ifconfig dummy0  
dummy0: flags=195<UP,BROADCAST,RUNNING,NOARP> mtu 1500  
ether 3a:4f:eb:62:41:5b txqueuelen 1000 (Ethernet)  
RX packets 0 bytes 0 (0.0 B)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 3 bytes 972 (972.0 B)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
abdo@abdo:~$ nmcli con show lab-net  
connection.id: lab-net  
connection.uuid: 3cf1b663-05f8-4741-a297-af18ceebf9e4  
connection.stable-id: --  
connection.type: dummy  
connection.interface-name: dummy0  
connection.autoconnect: yes  
connection.autoconnect-priority: 0  
connection.autoconnect-retries: -1 (default)  
connection.multi-connect: 0 (default)  
connection.auth-retries: -1  
connection.timestamp: 1767878556  
connection.read-only: no  
connection.permissions: --  
connection.zone: --
```

26. Reconfigure your network card using system-config-network utility to have static IP

```
abdo@abdo:~$ system-config-network-tui
system-config-network-tui: command not found
abdo@abdo:~$
```

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27. Configure your network card to have 3 IPs and check that they are all working using ifconfig command.

```
abdo@abdo: ~
abdo@abdo:~$ nmcli con mod lab-net ipv4.addresses 10.10.10.10/24
nmcli con mod lab-net +ipv4.addresses 10.10.10.11/24
nmcli con mod lab-net +ipv4.addresses 10.10.10.12/24
nmcli con up lab-net
Error: Connection activation failed: IP configuration could not be reserved (no available address, timeout, etc.)
Hint: use 'journalctl -xe NM_CONNECTION=3cf1b663-05f8-4741-a297-af18ceebf9e4 + NM_DEVICE=dummv0' to get more details.
abdo@abdo:~$ ifconfig dummy0
dummy0: flags=195<UP,BROADCAST,RUNNING,NOARP> mtu 1500
    inet 10.10.10.10 netmask 255.255.255.0 broadcast 10.10.10.255
    ether 3a:4f:eb:62:41:5b txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14 bytes 1998 (1.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

abdo@abdo:~$
```


28. Change your host name in your global network file.

```
abdo@abdo: ~  
abdo@abdo:~$ sudo hostnamectl set-hostname ahmed  
[sudo] password for abdo:  
abdo@abdo:~$ hostnamectl status  
Static hostname: ahmed  
Icon name: computer-laptop  
Chassis: laptop  
Machine ID: e4f23fdd13ea4253b5116005f45af51a  
Boot ID: ffd22e34fa0740368a13cbdf467238a3  
Operating System: Ubuntu 22.04.5 LTS  
Kernel: Linux 6.8.0-90-generic  
Architecture: x86-64  
Hardware Vendor: Dell Inc.  
Hardware Model: G3 3579  
abdo@abdo:~$
```

Lab2

1. Change your default SELinux mode to permissive and reboot.

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```
sudo vi /etc/selinux/config
```

```
SELINUX=enforcing → SELINUX=permissive
```

```
sudo reboot
```

2. After reboot, verify the system is in permissive mode.

```
getenforce
```

3. Change the default SELinux mode to enforcing.

```
SELINUX=enforcing
```


4. Change the current SELinux mode to enforcing.

```
sudo setenforce 1
```

```
getenforce
```

5. Copy /etc/resolv.conf file to root's home directory.

```
cp /etc/resolv.conf /root/
```

6. Observe the SELinux context of the initial /etc/resolv.conf

7. Move resolv.conf from root's home directory to /etc/resolv.conf

8. Observe the SELinux of the newly copied /etc/resolv.conf

```
abdo@abdo:~$ cp /etc/resolv.conf /root/
cp: cannot stat '/root/resolv.conf': Permission denied
abdo@abdo:~$ sudo cp /etc/resolv.conf /root/
abdo@abdo:~$ ls -Z /etc/resolv.conf
error: Found argument '-Z' which wasn't expected, or isn't valid in this context

USAGE:
    lsd [FLAGS] [OPTIONS] [--] [FILE]...

For more information try --help
abdo@abdo:~$ nano ~/.bashrc
abdo@abdo:~$ source ~/.bashrc
abdo@abdo:~$ ls -Z /etc/resolv.conf
? /etc/resolv.conf
abdo@abdo:~$ mv /root/resolv.conf /etc/resolv.conf
mv: cannot stat '/root/resolv.conf': Permission denied
abdo@abdo:~$ sudo mv /root/resolv.conf /etc/resolv.conf
abdo@abdo:~$ ls -Z /etc/resolv.conf
? /etc/resolv.conf
abdo@abdo:~$
```

9. Restore the SELinux context of the newly positioned `/etc/resolv.conf`

```
sudo restorecon -v /etc/resolv.conf
```

10. Observe the SELinux context of the restored `/etc/resolv.conf`

```
ls -Z /etc/resolv.conf
```